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Comparing muscle coordination patterns in yoga-focused exercises with those in standard stabilization exercises

Hiroki Saito [a,b,c,*], Yumeka Iwai [a], Ayu Yamano [a], Kazuya Matsushita [b,d], Hikaru Yokoyama [e], Kimitaka Nakazawa [c], Sakiko Ito [a,b]

[a] Department of Rehabilitation, Tokyo University of Technology, Tokyo, Japan
[b] Center for Human Movement, Tokyo University of Technology, Tokyo, Japan
[c] Graduate School of Arts and Sciences, Department of Life Sciences, The University of Tokyo, Tokyo, Japan
[d] Aobadai Takeda Orthopedic, Tokyo, Japan
[e] Institute of Engineering, Tokyo University of Agriculture and Technology, Tokyo, Japan

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ABSTRACT

Background: Yoga-focused exercises are increasingly recognized as a potential tool for aiding individuals with musculoskeletal conditions, such as low back pain. This study aimed to investigate the differences in muscle synergies during yoga-focused exercises compared to standard stabilization exercises, to clarify the neuromechanical mechanisms underlying the use of yoga-focused exercises.
Methods: Eleven healthy males participated in the study. Unilateral surface EMG data were collected from 15 muscle groups in the trunk, upper, and lower limbs during both yoga-focused and standard exercises. Non-negative matrix factorization (NMF) was utilized to extract muscle synergies.
Results: Yoga-focused exercises demonstrated a significantly higher number of muscle synergies compared to standard stabilization exercises ($p = 0.02$, effect size = 1.09), with median values of 9.0 (range 8.0–11.0) for yoga-focused exercises and 8.0 (range 6.0–10.0) for standard exercises. Several yoga-specific muscle synergies were identified, highlighting a more diverse array of muscle activation patterns, particularly in the lumbopelvic region.
Conclusions: The distinct muscle synergies observed in yoga-focused exercises suggest a unique neuromechanical profile, potentially contributing to improved muscle coordination for lumbopelvic function.

1. Introduction

In the modern health and fitness paradigm, yoga, a practice deeply rooted in ancient Indian tradition, is gaining rapid recognition as a significant tool to enhance physical wellness and agility (Telles et al., 2021). These exercises are not merely physical postures, but an intricate blend of poses (asanas), controlled breathing (pranayama), and frequently, meditation (dhyana), fostering a synergy that is aimed at both mental clarity and physical vitality(Hayes and Chase, 2010). Over the years, particularly in the west, yoga has evolved into a widely adopted intervention, with over 21 million adults in the U.S. incorporating it into their wellness routines (Cramer et al., 2016).

Therapeutic yoga, a subset of the larger yoga discipline, is being increasingly recognized as a viable means to assist individuals in managing various health conditions and alleviating associated symptom (McCall et al., 2013). It has been particularly noted for its potential in managing pain and related disabilities across a range of conditions, prominently including persistent low back pain (LBP) (Bussing et al., 2012; T, 2007). Indeed, several randomized control trials (RCTs) showed superior effect of yoga-focused exercises on back specific function and pain when compared with standard stabilization exercises such as abdominal curl-up, back extensions and lower limb strengthening (Demirel et al., 2019; Nambi et al., 2014; Patil et al., 2018; Tekur et al., 2008).

Yoga-focused exercises, through the practice of various poses, are believed to enhance muscle strength, flexibility, and dynamic balance (Beazley et al., 2017; Krejci et al., 2022; Lehecka et al., 2021; Longpre et al., 2015) potentially offering benefits for lumbopelvic function in individuals with LBP. Previous studies using electromyography (EMG) have demonstrated the potential underlying mechanisms for these

* Corresponding author. Department of Rehabilitation, Tokyo University of Technology, Tokyo, Japan.
E-mail address: saitohhrk@stf.teu.ac.jp (H. Saito).

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specific benefits in persistent pain conditions. They reported that basic yoga postures, including plank and bridge, are associated with greater activity in trunk and hip muscles (Beazley et al., 2017; Bolgla et al., 2018). However, these studies were limited to selected yoga postures and relatively few muscle recordings, and they lacked a comparison of EMG data with that from standard stabilization exercises. Although both exercises share fundamental postures such as standing, squatting, and forward and backward bending, yoga-focused exercises are expected to involve more complex movement of the trunk, upper limbs, and lower limbs. Thus, the specificity of yoga-focused exercises as an effective therapeutic means can be attributed to more whole-body coordination patterns compared to standard stabilization exercises, which might benefit from an increase in movement tasks and muscle recordings to reveal these patterns.

Dimensionality reduction algorithms, such as non-negative matrix factorization (NMF), are beneficial for extracting underlying structures of coordinated muscle patterns, termed muscle synergies, from the covariability in electromyography (EMG) data of multiple muscles during human behaviors (d'Avella and Bizzi, 2005; d'Avella et al., 2003). Through a detailed analysis, we aim to unveil the muscle coordination strategies during yoga-focused exercises when compared to standard exercises in health individuals. We hypothesized that the number of muscle synergies is higher in yoga than in standard exercises and that several yoga-specific synergies are extracted, potentially enhancing their efficacy in treating LBP.

2. Methods

2.1. Participants

Eleven healthy individuals, with a mean age (±SD) of 21.0 ± 0.6 years, were recruited from a local university. The mean height and weight of the participants were 169.0 ± 8.2 cm and 61.2 ± 10.2 kg, respectively. Each participant provided written informed consent to participate in the study. The study was conducted in accordance with the principles of the Declaration of Helsinki and was approved by the local ethics committee of the University of Tokyo (762).

2.2. Experimental procedures

Participants were asked to perform 20 yoga-focused exercises and 12 standard exercises, as shown in Fig. 1. Yoga-focused exercises were selected based on previous studies with RCTs (Saper et al., 2009, 2013). Since we were interested in movement complexity across the whole body, we excluded yoga tasks that involved minimal movement where no or minimal muscle activations were expected. Standard exercises, also based on previous studies with RCTs (Demirel et al., 2019; Neyaz et al., 2019; Patil et al., 2018) were selected, excluding stretching exercises where no or minimal muscle activations were expected. Each task was repeated three times and the order of the tasks was randomly assigned. A yoga practitioner with five years of experience provided verbal instructions and demonstrated each task.

2.3. Data collection

Unilateral surface EMG data were recorded from 15 lower limb and trunk muscle groups: rectus abdominis (RAS) (3 cm lateral to umbilicus) (McGill et al., 1996), oblique externus (OE) (15 cm lateral to umbilicus) (Vera-Garcia et al., 2011), erector spinae at Th9 (ESL1) (3 cm lateral to the TH9 spinous process) (McGill et al., 1996), Latissimus dorsi (LD), gluteus maximus (GM), gluteus medius (Gmed), adductor longus (ADD), biceps femoris (long head, BF), rectus femoris (RF), gastrocnemius medialis (MG), tibialis anterior (TA), Middle Trapezius (MT), Deltoid Middle (DM), long head of the biceps (BCL) and long head of the triceps (TLo). The EMG sensor placements in lower limb were based on SENIAM (surface EMG for a non-invasive assessment of muscles) (Hermens et al., 2000). A wireless EMG system (Trigno Wireless System; DELSYS, Boston, MA, USA) was used to record EMG activity. The EMG signals were band-pass filtered (20–450 Hz), amplified (with a 300-gain preamplifier), and sampled at 2000 Hz using an analog-to-digital converter (Power lab/16SP, AD Instruments, Australia).

2.4. EMG processing

Raw EMG signals were high-pass filtered at 30 Hz to remove motion artifacts, and then demeaned (Redfern et al., 1993). The signals were then full-wave-rectified and low-pass filtered at 10 Hz, using a fourth-order Butterworth filter (Turpin et al., 2021). The smoothed EMG envelopes were time-interpolated to generate 100 time points between the start and end points for each trial so that the EMG data of each trial contributed to the extracted muscle synergies equally(Saito et al., 2023b).

We created all-task EMG matrices for each exercise for each participant. Specifically, the matrix was composed of 15 muscles and 6000 timepoints for yoga-focused exercises and 3600 timepoint for standard exercises, derived from the number of tasks multiplied by the number of repetitions/cycles (3) and 100 samples, to extract the muscle synergies for each exercise. EMG from each muscle was normalized to the

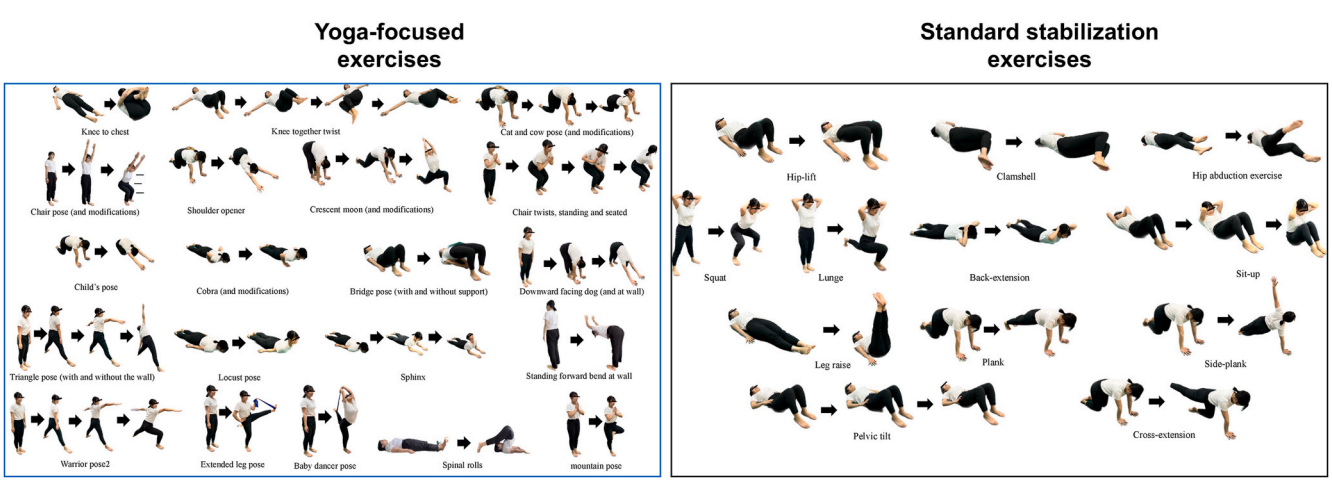

Fig. 1. Visual descriptions of yoga-focused exercises and standard exercises. Each participant were asked to perform 20 yoga-focused exercises and 12 standard exercises. Each task was repeated three times and the order of the tasks was randomly assigned. Five years of experienced yoga practitioner gave a verbal description and demonstrated tasks.

maximum amplitude across all trials.

2.5. Independent muscle synergy extraction

To extract muscle synergies, non-negative matrix factorization (NMF) was implemented using MATLAB R2023b (MathWorks Inc., Natick, MA, USA), and applied to the all-task EMG matrix for each exercise. NMF has previously been described as a linear decomposition technique (Lee and Seung, 1999; Tresch et al., 2006) according to equation (1):

$$M = W \cdot C + e \quad (1)$$

where M ($m \times t$ matrix, where m is the number of muscles, t is the number of samples [i.e., spatiotemporal profiles of muscle activity]) is a linear combination of muscle synergies, W ($m \times n$ matrix, where n is the number of muscle synergies), C ($n \times t$ matrix, representing temporal patterns), and e is the residual error matrix. We applied NMF to extract possible n values from 1 to 15 for each dataset. To estimate the optimal number of muscle synergies, the variance accounted for (VAF) by the reconstructed EMG (M) was calculated at each iteration (Torres-Oviedo et al., 2006). VAF was defined as 100 × the square of the uncentered Pearson's correlation coefficient (Torres-Oviedo et al., 2006; Zar, 1999). Each synergy extraction was repeated 10 times, and the VAF was calculated at each iteration. Iterations with the highest VAF were maintained (Saito et al., 2021b, 2022; Yokoyama et al., 2016, 2021). VAFs >0.9 were used to identify the optimal number of synergies commonly used in the literature (Botzheim et al., 2021; Hug et al., 2011; Saito et al., 2021b, 2022; Sawers et al., 2015; Stamenkovic et al., 2021).

2.6. Representative shared and specific synergies

We identified representative shared and specific synergies across participants using hierarchical clustering analysis (Ward's method, Euclidian distance) of muscle weighting components (Yokoyama et al., 2016). To determine the optimal number of clusters, we computed the gap statistic (Tibshirani et al., 2001), which measures the compactness of the clustering achieved against those in reference data sets without any obvious clustering similar to a previous study (Cheung et al., 2020). Reference data sets (N = 500) were initially generated by sampling uniformly from within the bounds of the original muscle-synergy set; each of them was then clustered by the hierarchical cluster, at 2–20 clusters. The optimal number of clusters was then the smallest number, h, such that

$$\text{Gap (h)} \geq \text{Gap (h+1)} - \text{sd (h+1)}$$

where Gap(k) represents the gap statistic at h clusters, and sd(h) signifies the standard deviation of the clustering compactness within the reference data sets (Tibshirani et al., 2001).

We focused the comparison of muscle weighting components between yoga-focused exercises and standard exercises to explore the relative contribution of muscles for each exercise. In this context, we calculate the similarity defined by scalar products between centroids of the clusters in muscle weighting components normalized by unit vectors (Cheung et al., 2020). The synergy clusters that could not be matched with scalar product ≥0.8 were classified as unmatched (Cheung et al., 2020).

2.7. Statistics

We compared the number of muscle synergies between exercises using a non-parametric Wilcoxon signed-rank test was used due to the small sample size and non-normal distribution assumptions. When there was a significant difference between the groups, effect sizes (ES) were calculated using Cohen's d (Cohen, 1988). We recruited eleven participants without an a priori power analysis. Thus, we instead conducted a sensitivity analysis in G*Power, which indicated that an effect size of 0.93 would be necessary to obtain a power of 80 % at an α of 0.05.

3. Results

3.1. EMG patterns of yoga-focused exercises and standard exercises

Fig. 2 shows the concatenated EMG envelops of yoga-focused exercises and standard exercises. The mean of the EMG envelopes for both exercises are plotted as a line and the standard deviation as a shading around it. Based on visual inspections, there were no clear artifacts in the recorded EMG signals both tasks and overall.

3.2. Muscle synergies in yoga-focused exercises and standard exercises

Fig. 3 (a) shows the VAF for each muscle synergy in yoga-focused exercises and standard exercises. As shown in Fig. 3 (b), the median values of extracted number of synergies were 9.0 (8.0–11.0) in yoga-focused exercises and 8.0 (6.0–10.0) in standard exercises. Yoga-focused exercises showed a significantly higher number of muscle synergies compared with those in standard exercises ($p = 0.02$, ES = 1.09). This observed effect size exceeds the threshold identified in our G*Power sensitivity analysis (0.93), indicating that the sample size was sufficient to detect the observed difference.

Fig. 4 presents the centroids of clusters in muscle weighting components using the cluster analysis. We identified eleven clusters in yoga-focused exercises and eight clusters in standard exercises. Synergy 1 to 8 were highly similar between exercises (SP > 0.90) while Synergy 9–10 were unmatched and thus classified as yoga-specific muscle synergies.

4. Discussion

This study aimed to compare muscle synergies, with a special focus on muscle weighting components, between yoga-focused exercises and standard stabilization exercises in healthy individuals. The goal was to comprehend the neuromechanical mechanisms that distinguish yoga-focused exercises as potential therapeutic interventions compared with standard stabilization exercises. While a large proportion of muscle synergies were shared between the two types of exercises, our results revealed that the number of muscle synergies in yoga-focused exercises was significantly higher than that in standard exercises, with several yoga-specific synergies identified. These findings indicate that yoga-focused exercises require higher neuromechanical demands, accompanied by a diverse array of muscle activation patterns.

The high similarity of muscle synergies between yoga-focused exercises and standard stabilization exercises indicates that these tasks share a foundational neuromuscular control strategy, marked by the orchestrated activation of upper, lower limb, and trunk muscles. This phenomenon of shared synergies across distinct behaviors is well-documented in earlier studies (Barroso et al., 2014; Chvatal et al., 2011; Sawers et al., 2015). Both exercises revealed functionally minimal units of muscle synergies, such as Synergy 1—comprising RAS and OE activation patterns contributing to trunk movement and stability(Saito et al., 2023a), and Synergy 2, involving the activation of ESTH9, LD, and MT, pivotal for back extension (Saito et al., 2023a). These minimal units represent sparser forms of muscle synergy. When these synergies are sequentially burst and/or merged, they are advantageous for executing a diverse range of movements and postures that necessitate both the acceleration/deceleration and stability of specific joints (Saito et al., 2021b, 2022). For instance, the activation of Synergy 2, followed by Synergy 1, results in trunk backward and return movement. Conversely, when tasks require stability, multiple minimal units of muscle synergies may merge or activate simultaneously to intensify the co-contractions of multiple muscles to counterbalance gravity or to stiffen the specific joints, such as the co-activations of Synergy 1 and Synergy 2 to produce spinal stability.

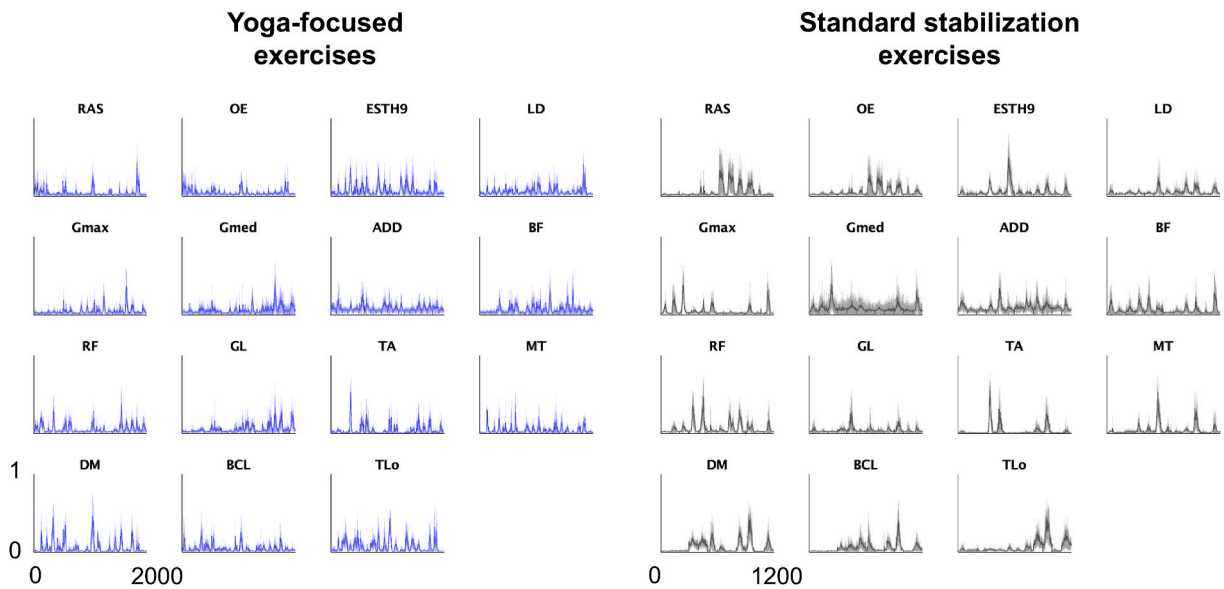

Fig. 2. Muscle activation patterns during yoga-focused exercises and standard exercises. The combined electromyography (EMG) envelope of three repetitions of tasks for each exercise is shown. The mean of the EMG envelopes for all eleven participants is represented as a line, with the standard deviation shown as shading around it. The amplitude is normalized to the maximum value for each muscle across all tasks.

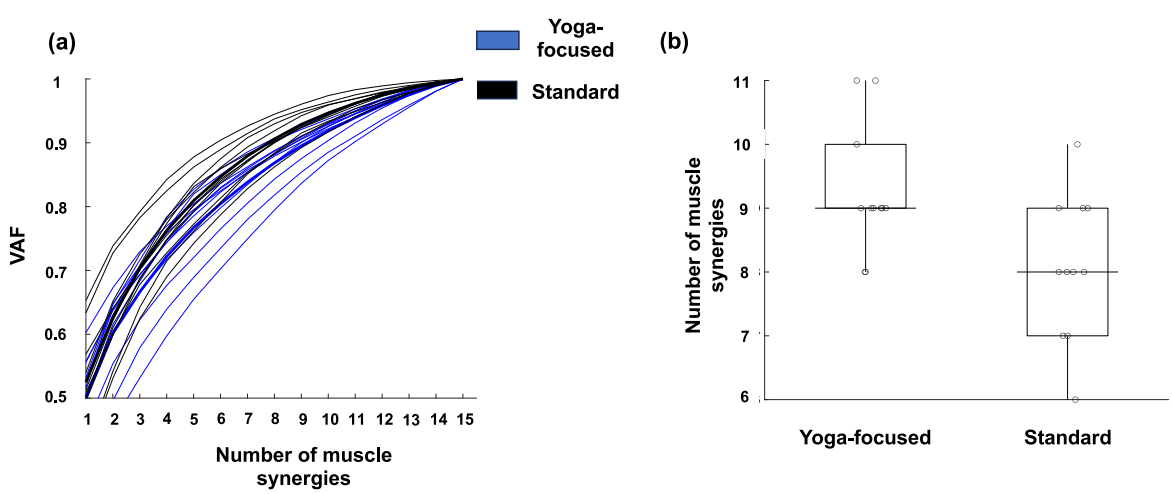

Fig. 3. (a) Variance accounted for (VAF) by each muscle synergy during yoga-focused exercises and standard exercises. Individual (thin lines) and overall participant (thick lines) means of VAF are presented for each task. (b) The number of muscle synergies between yoga-focused exercises and standard exercises. The box edges represent the 25th and 75th percentiles. The median values of extracted number of synergies were 9.0 (8.0–11.0) in yoga-focused exercises and 8.0 (6.0–10.0) in standard exercises. Yoga-focused exercises showed the significantly higher number of muscle synergies compared with those in standard exercises ($p = 0.02$, ES $= 1.09$).

In addition to the shared foundation of neuromechanics in both exercises, yoga-focused exercises were found to involve a higher number of muscle synergies compared to standard stabilization exercises, identifying three yoga-specific synergies critical for lumbopelvic posture and movement control (Synergy 9–11). These include ESTH9 and LD patterns (Synergy 9), Gmed pattern (Synergy 10), and BF pattern (Synergy 11). Thus, these back and hip muscle patterns specific in yoga-focused exercises may additionally impact the function of specific muscles, enhancing stability, and flexibility in the lumbopelvic region for individuals with LBP. Furthermore, since the number of muscle synergies often mirrors the mechanics of a motor task, yoga-focused exercises necessitate a broader array of postures with more complex muscle activation patterns. Previous studies have demonstrated that reduced movement variability, potentially arising from the co-contraction of trunk and lower limb muscles, is linked to the onset and persistence of LBP (Abu Bakar et al., 2023; Alsubaie et al., 2023; Saito et al., 2021a). Consequently, exercises characterized by high movement variability are effective in enhancing optimal variability and alleviating LBP (Bauer et al., 2019). Our study intimates that yoga-focused exercises, with their higher number of muscle synergies, can offer a richer variety of muscle activation patterns compared with standard exercises, allowing individuals with LBP to adapt to a range of potentially threatening activities by accessing a substantial pool of muscle synergies and thus broadening the spectrum of feasible solutions to the mechanical demands of a task (Ting et al., 2015). Validation of this hypothesis necessitates future research.

This study presents several notable strengths. First, it utilized NMF for EMG analysis, a robust technique for extracting underlying muscle synergy patterns. Second, EMG signals were comprehensively recorded from 15 muscles spanning the trunk and both upper and lower limbs, providing a broad and detailed overview of muscle activation. This extensive muscle coverage enabled the identification of distinct

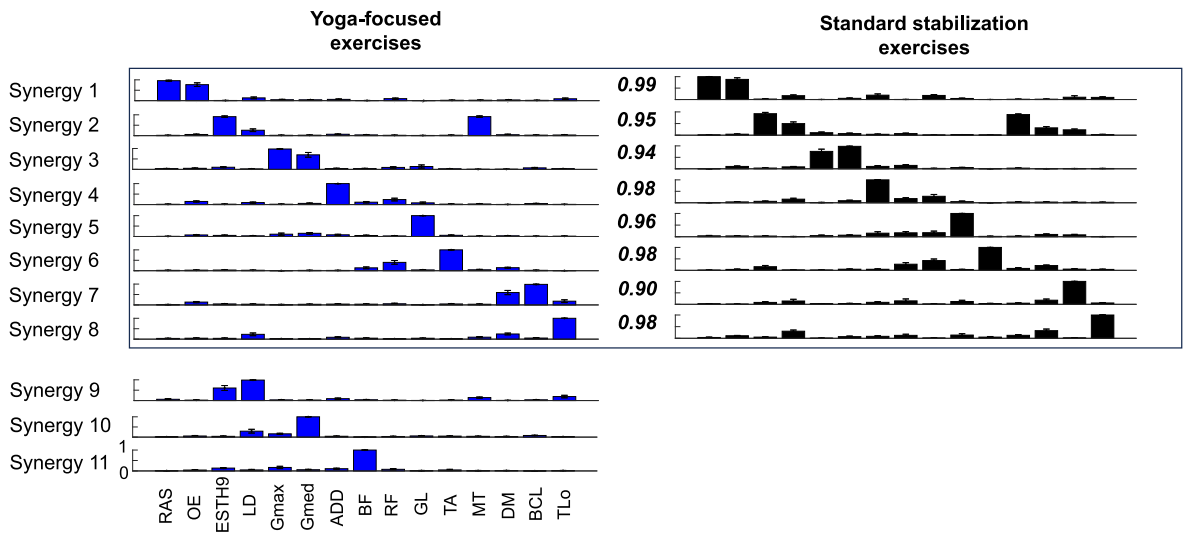

Fig. 4. Representative shared and specific muscle synergies identified by cluster analysis for muscle weighting components. Synergy 1–8 were shared between exercises. Synergy 9–10 were considered as yoga-specific synergy. RAS: Rectus Abdominis, OE: Obliquus Externus, ESTH9: Erector Spinae at TH9, GM: Gluteus Maximus, Gmed: Gluteus Medius, ADD: Adductor Longus, BF: Biceps Femoris, RF: Rectus Femoris, ML: Gastrocnemius Lateralis, TA: Tibialis Anterior, MT: Middle Trapezius, DM: Deltoid Middle, BCL: Biceps Longus, and TLo: Triceps Longus.

coordination patterns, including yoga-specific muscle synergies not observed in standard stabilization exercises. Despite these strengths, several limitations should be acknowledged. The sample size was relatively small ($n = 11$) and consisted only of healthy young male participants, which restricts the generalizability of the findings to other populations (e.g., females, older adults, or individuals with musculoskeletal conditions). In addition, the study did not include clinical outcome measures such as pain levels or functional performance, so it remains uncertain whether the observed differences in muscle coordination translate into meaningful clinical benefits.

5. Conclusion

This study elucidated the neuromechanical differences between yoga-focused and standard exercises. The findings indicate that yoga-focused exercises manifest a more substantial number of muscle synergies, leading to a multifaceted spectrum of muscle activation patterns. Such complexity in yoga-focused exercises may propose a therapeutic edge, providing an expansive pool of muscle synergies to facilitate a variety of muscle activation patterns and potentially improve lumbopelvic control. However, these findings are limited by the small sample size. Future studies should confirm these results in larger and more diverse populations to improve generalizability.

CRediT authorship contribution statement

Hiroki Saito: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yumeka Iwai:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Data curation, Conceptualization. **Ayu Yamano:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Data curation. **Kazuya Matsushita:** Writing – review & editing, Software, Methodology, Investigation, Data curation, Conceptualization. **Hikaru Yokoyama:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Kimitaka Nakazawa:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Sakiko Ito:** Writing – review & editing, Validation, Methodology, Investigation, Conceptualization.

Ethics approval and consent to participate

All experimental protocols were approved by the local ethics committee of XX, and all participants gave their written informed consent.

Consent for publication

The participant depicted in the photos gave their written consent for publication.

Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Permission to reproduce material from other sources

N/A.

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Declaration of competing interest

None declared.

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